

FPGA Synthesis and Prototyping Lab:

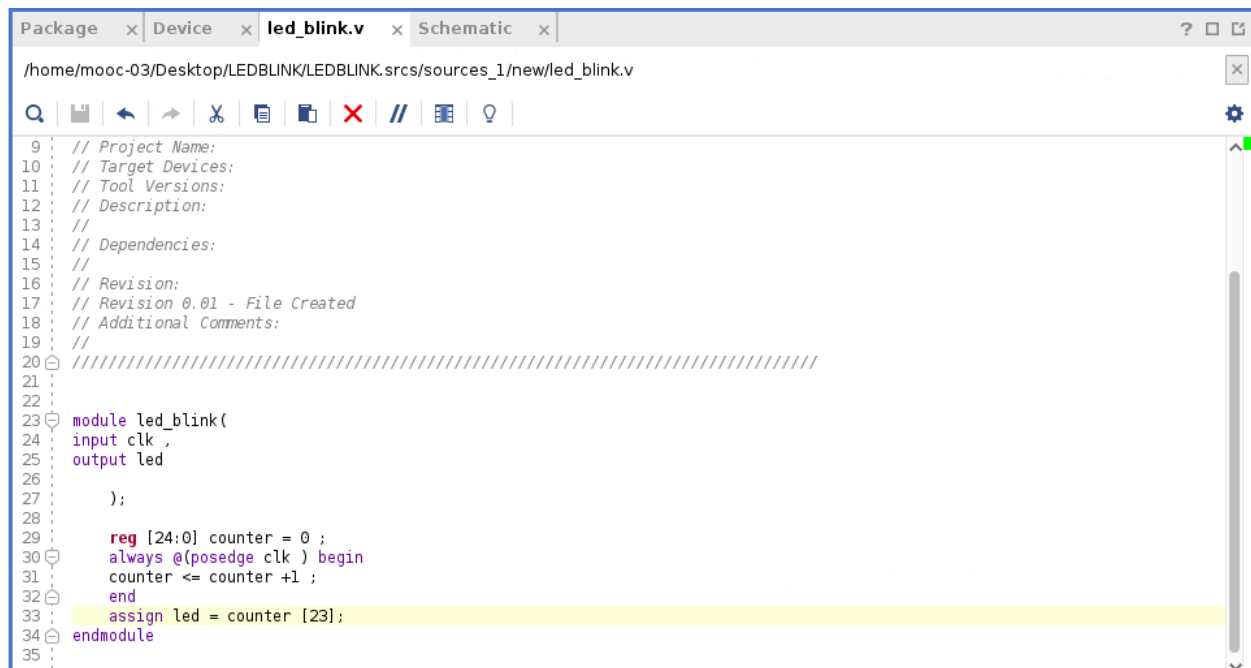
34

Basic Gates

Introduction

The purpose of this experiment is to Synthesis and Check the behavior of several of the basic logic gates mentioned in Lab-1

CODE:



```
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module led_blink(
24     input clk ,
25     output led
26
27 );
28
29     reg [24:0] counter = 0 ;
30     always @(posedge clk ) begin
31         counter <= counter +1 ;
32     end
33     assign led = counter [23];
34 endmodule
35
```

CONSTRAINTS

RTL Schematic

Package x Device x led_blink.v x Schematic x cons.xdc x

/home/mooc-03/Desktop/LEDBLINK/LEDBLINK.srcs/constrs_1/new/cons.xdc

1 set_property PACKAGE_PIN E3 [get_ports clk]
2 set_property PACKAGE_PIN H5 [get_ports led]
3 set_property IOSTANDARD LVCMOS33 [get_ports clk]
4 set_property IOSTANDARD LVCMOS33 [get_ports led]
5

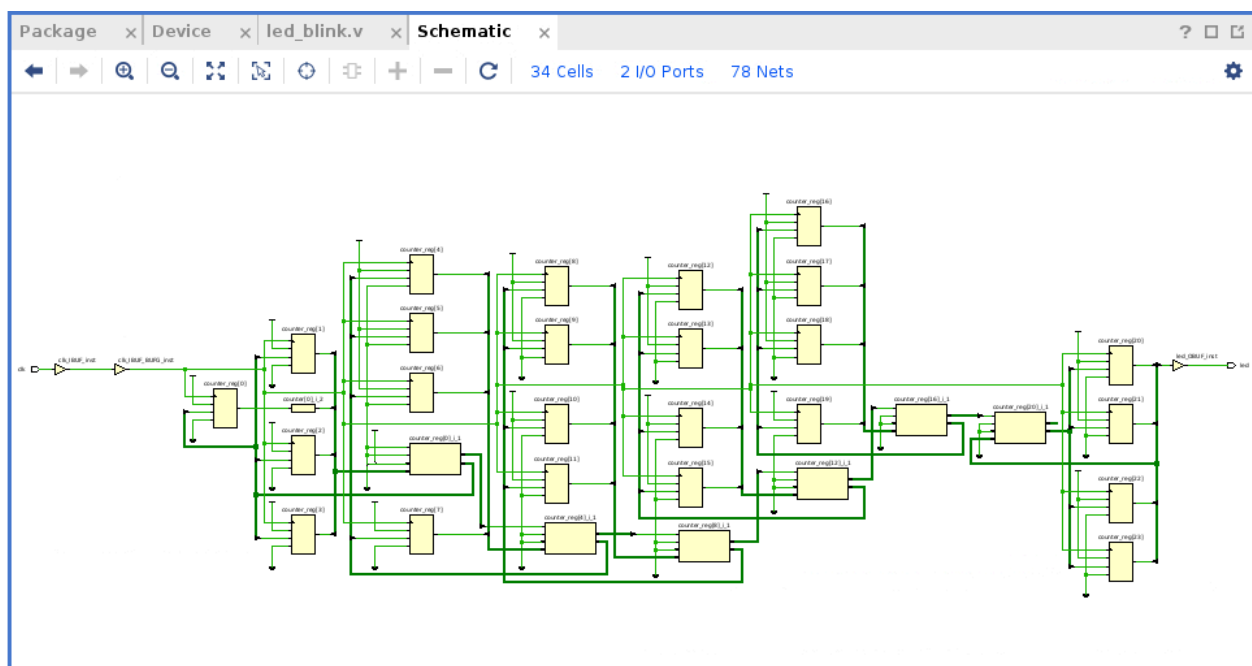
Project Summary x Schematic x led_blink.v x cons.xdc x Schematic (2) x

26 Cells 2 I/O Ports 52 Nets

Tcl Console Messages Log Reports Design Runs x

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodol
synth_1	constrs_1	synth_design Complete!									
impl_1	constrs_1	write_bitstream Complete!	NA	NA	NA	NA		NA	0.916	0	24 CW

Synthesis Report



Project Summary x Schematic x led_blink.v x cons.xdc x Schematic (2) x

Overview | Dashboard

Settings Edit

Project name: LEDBLINK

Project location: /home/mooc-03/Desktop/LEDBLINK

Product family: Artix-7

Project part: xc7a100tcsq324-1

Top module name: led_blink

Target language: Verilog

Simulator language: Mixed

Synthesis

Status: ✓ Complete

Messages: ! 2 warnings

Part: xc7a100tcsq324-1

Strategy: [Vivado Synthesis Defaults](#)

Report Strategy: [Vivado Synthesis Default Reports](#)

Incremental synthesis: [Automatically selected checkpoint](#)

Implementation

Status: ✓ write_bitstream Complete!

Messages: ! 12 warnings

Part: xc7a100tcsq324-1

Strategy: [Vivado Implementation Defaults](#)

Report Strategy: [Vivado Implementation Default Reports](#)

Incremental implementation: [None](#)

DRC Violations

Summary: ! 1 warning

[Implemented DRC Report](#)

Timing

Worst Negative Slack (WNS): NA

Total Negative Slack (TNS): NA

Number of Failing Endpoints: NA

Total Number of Endpoints: NA

[Implemented Timing Report](#)

Utilization

Post-Synthesis | Post-Implementation

Graph | Table

LUT 1%

FF 1%

IO 1%

BUFG 3%

Power

Summary | On-Chip

Total On-Chip Power: 0.916 W

Junction Temperature: 29.2 °C

Thermal Margin: 55.8 °C (12.1 W)

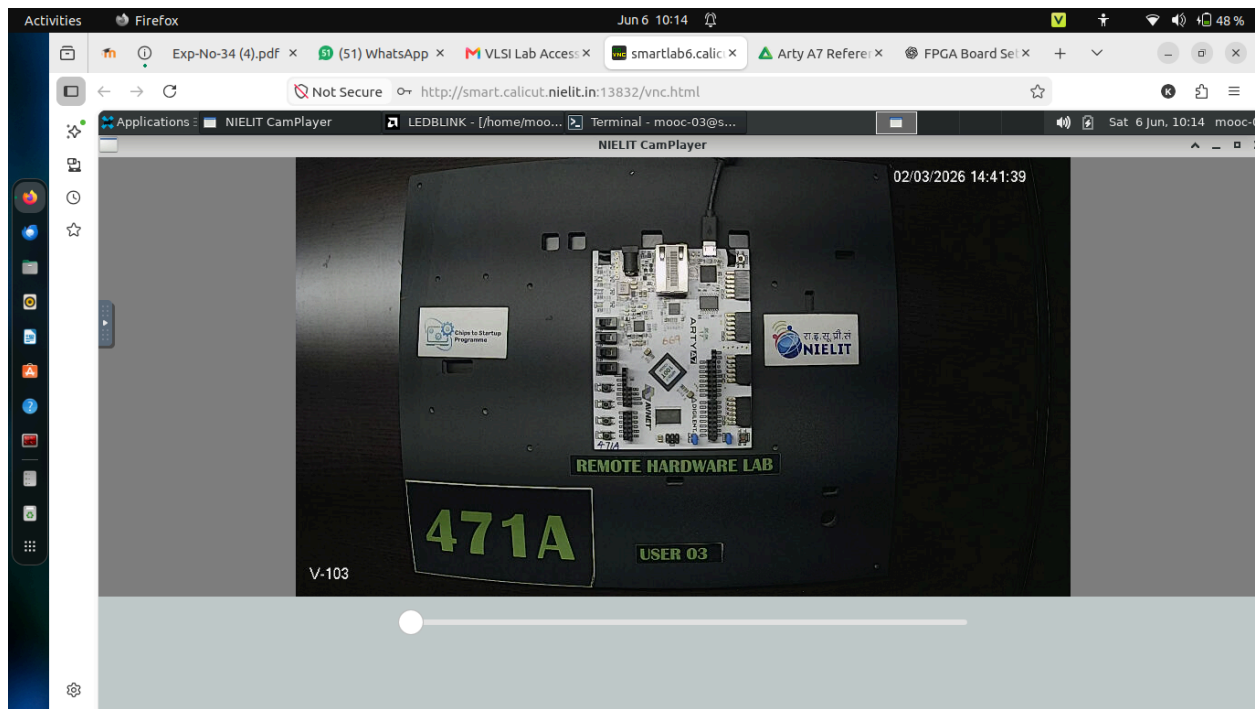
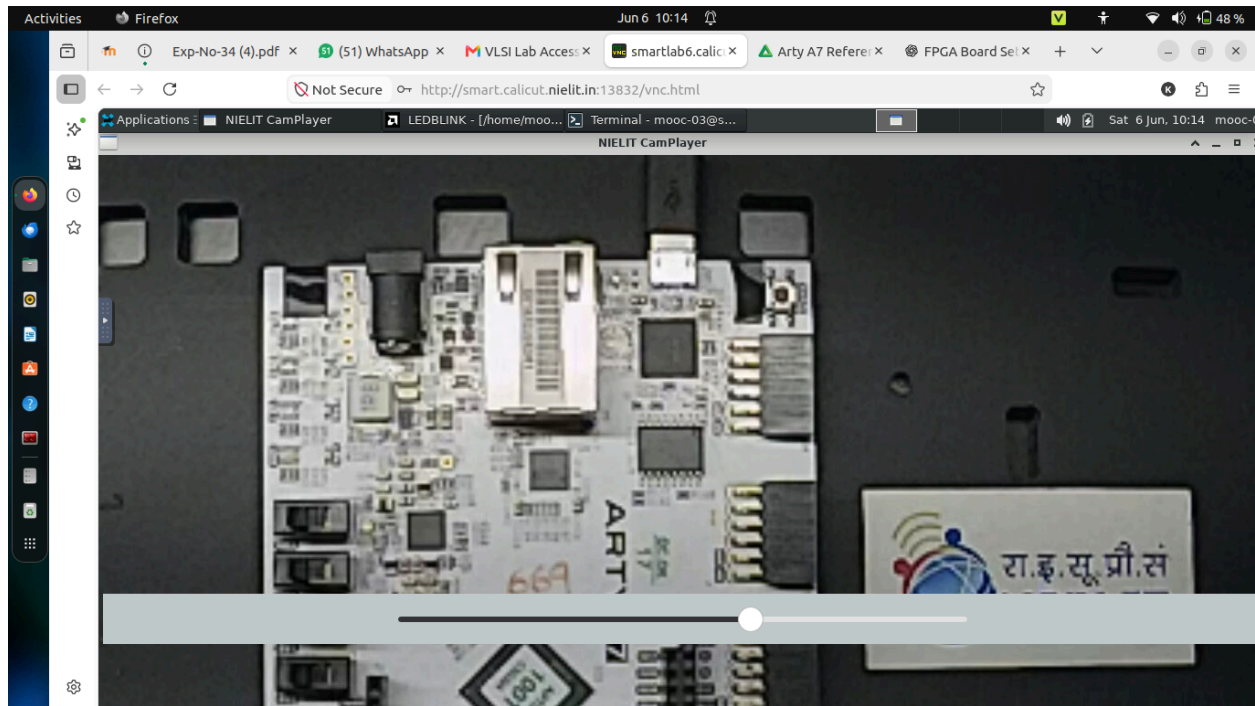
Effective θ_{JA} : 4.6 °C/W

Power supplied to off-chip devices: 0 W

Confidence level: [Low](#)

[Implemented Power Report](#)

OUTPUTS
 LED IS BLINKING ;



Consol

```
7 Infos, 0 Warnings, 0 Critical Warnings and 0 Errors encountered.
synth_design completed successfully
synth_design: Time (s): cpu = 00:00:10 ; elapsed = 00:00:06 . Memory (MB): peak = 9952.750 ; ga
INFO: [Common 17-2834] synth_design peak Physical Memory [PSS] (MB): overall = 3235.341; main =
INFO: [Common 17-2834] synth_design peak Virtual Memory [VSS] (MB): overall = 9942.715; main =
set_property PROBES.FILE {} [get_hw_devices xc7a100t_0]
set_property FULL_PROBES.FILE {} [get_hw_devices xc7a100t_0]
set_property PROGRAM.FILE {/home/mooc-03/Desktop/LEDBLINK/LEDBLINK.runs/impl_1/led_blink.bit} [
program_hw_devices [get_hw_devices xc7a100t_0]
INFO: [Labtools 27-3164] End of startup status: HIGH
refresh_hw_device [lindex [get_hw_devices xc7a100t_0] 0]
INFO: [Labtools 27-1434] Device xc7a100t (JTAG device index = 0) is programmed with a design th
```